

DAY-4

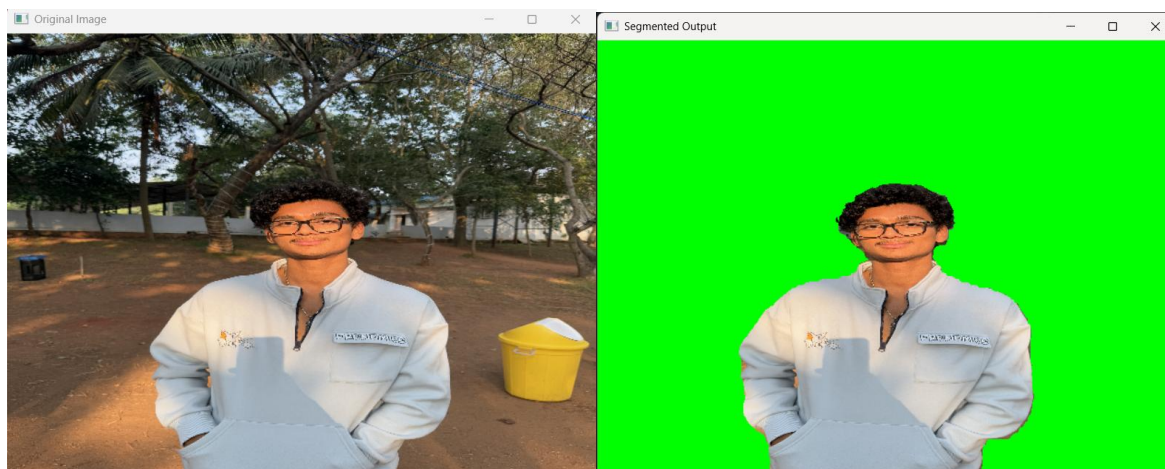
TASK 3: TESTING BODY SEGMENTATION METHODS:

I learned to implement body segmentation using OpenCV and MediaPipe itself with the help of this code.

```
segmentation.py > ...
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  mp_selfie_segmentation = mp.solutions.selfie_segmentation
5  segment = mp_selfie_segmentation.SelfieSegmentation(model_selection=1)
6  image = cv2.imread("pran.JPG")
7  image = cv2.resize(image, (640, 480))
8  rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
9  results = segment.process(rgb)
10 mask = results.segmentation_mask
11 condition = mask > 0.5
12 bg_color = (0, 255, 0)
13 bg_image = image.copy()
14 bg_image[:] = bg_color
15
16 output = np.where(condition[..., None], image, bg_image)
17 cv2.imshow("Original Image", image)
18 cv2.imshow("Segmented Output", output)
19
20 cv2.waitKey(0)
21 cv2.destroyAllWindows()
```

Separated the human body from the background successfully.

OUTPUT:



GIVEN IMAGE

SEGMENTED IMAGE

IMPORTANCE:

Body segmentation is important because it helps AI accurately identify and isolate the human body from the background. In virtual try-on systems, this allows clothes to be properly aligned and overlaid on the user without affecting the background area.

It also improves body measurement accuracy, cloth fitting, and overall visual realism. Segmentation is widely used in applications such as virtual fashion, video conferencing background removal, fitness tracking, AR filters, and image editing tools.

WHAT I LEARNED:

While working on body segmentation, I learned how AI can separate a person from the background using computer vision techniques.

I implemented segmentation using MediaPipe and OpenCV, where the model creates a mask around the human body for accurate detection.

I also understood how threshold values, lighting conditions, and image quality affect segmentation accuracy. This gave me a better understanding of how virtual try-on systems and background removal applications work in real-world AI solutions.